

Self-Organizing Logos (SOL): Mathematical Foundation v2

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1. Notation and Conceptual Mapping

Let $(\mathcal{M}, g)$ be a low-dimensional Riemannian manifold serving as a conceptual state-space.

- $x \in \mathcal{M}$ is a conceptual “location” (state).
- $t \geq 0$ is interaction time.
- g is the Riemannian metric inducing gradient $\nabla_{\mathcal{M}}$ and divergence $\nabla_{\mathcal{M}}$.

We define the SOL semantic state field

$$u(x, t) = (\rho(x, t), v(x, t), \phi(x, t)), \quad (1)$$

where:

- $\rho(x, t) \geq 0$ is semantic mass / activation density,
- $v(x, t) \in T_x \mathcal{M}$ is a semantic flow (velocity) field,
- $\phi(x, t) \in \mathbb{R}$ is a semantic potential (a structured “drive” that can shape flow).

We also track semantic modes (topics/roles/perspectives) via

$$\psi(x, t) \in \mathbb{R}^k. \quad (2)$$

1.1 Notation Summary

Symbol	Meaning
$\mathcal{M}$	Conceptual manifold (semantic state-space)
g	Riemannian metric on $\mathcal{M}$
$x \in \mathcal{M}$	Conceptual location
$t \geq 0$	Interaction time
$u(x, t) = (\rho, v, \phi)$	Semantic state field
$\rho(x, t)$	Semantic mass / activation density
$v(x, t)$	Semantic flow (tangent vector field)
$\phi(x, t)$	Semantic potential
$p(x, t)$	Semantic pressure / Information Density Constraint
$\Pi(\cdot)$	Equation of state mapping density to pressure
$\psi(x, t) \in \mathbb{R}^k$	Semantic mode vector
K	Diffusion tensor / operator for ψ
$R(\psi; \cdot)$	Reaction term for ψ (node-wise / local)

2. Governing Equations on the Manifold

2.1 Continuity (Mass Conservation)

Semantic density evolves via a continuity equation:

$$\partial_t \rho + \nabla_{\mathcal{M}} \cdot (\rho v) = s_\rho(x, t), \quad (3)$$

where s_ρ is an optional source/sink capturing external stimulation (queries, observations, or injected attention).

2.2 Momentum (Inertia + Forcing + Regulation)

We introduce a semantic pressure field $p : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}$ interpreted as an Information Density Constraint.

Definition (Information Density Constraint). When too many concepts crowd into the same semantic region (high $\rho(x, t)$), internal pressure rises and forces flow to diverge toward other regions. We posit a monotone equation of state

$$p(x, t) = \Pi(\rho(x, t); \psi(x, t)), \quad \partial_\rho \Pi(\rho; \psi) \geq 0. \quad (4)$$

A practical default (stable, smooth, monotone) is, for constants $c > 0$ and $\rho_0 > 0$,

$$\Pi_0(\rho) = c \log(1 + \rho/\rho_0), \quad (5)$$

or, in “stiffer” regimes, a porous-medium style law $\Pi_0(\rho) = c\rho^\gamma$ with $\gamma > 1$.

We allow modes ψ to *tilt* regulation (topic/role-dependent capacity) via, e.g.,

$$\Pi(\rho; \psi) = \Pi_0(\rho) + \beta^\top \psi, \quad (6)$$

with $\beta \in \mathbb{R}^k$.

The momentum equation is modeled after compressible flow with optional damping/viscosity for stability:

$$\rho \left(\partial_t v + (v \cdot \nabla_{\mathcal{M}}) v \right) = -\nabla_{\mathcal{M}} p - \alpha \rho v + \mu \Delta_{\mathcal{M}} v + f(x, t; \psi), \quad (7)$$

where $\alpha \geq 0$ is a friction term, $\mu \geq 0$ is a (semantic) viscosity, $\Delta_{\mathcal{M}}$ is the Laplace–Beltrami operator, and f is an external forcing induced by inputs or controls. Figure 1 provides a schematic illustration of how high-density regions induce semantic pressure and deflect the flow field.

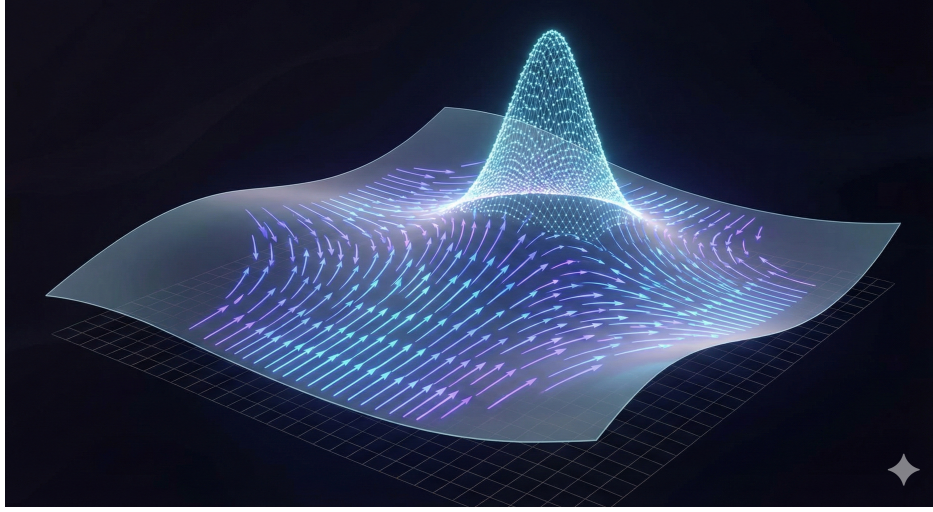


Figure 1: Conceptual visualization of the Riemannian manifold $(\mathcal{M}, g)$. A high-density conceptual cluster (the glowing peak) creates a region of high semantic pressure. The semantic flow vector field $v(x, t)$, represented by the arrows, is shown advecting (diverging and curving) around this high-pressure zone, rather than passing directly through it, illustrating the Information Density Constraint.

2.3 Conservative vs. Circulatory Flow (Resolving “Potential Flow” vs “Vortices”)

To retain the interpretability of a potential ϕ and still allow circulation (vortices/eddies as structured recurrence), we decompose:

$$v = -\nabla_{\mathcal{M}} \phi + w, \quad \nabla_{\mathcal{M}} \cdot w = 0. \quad (8)$$

Here $-\nabla_{\mathcal{M}} \phi$ is the conservative (gradient-driven) component, while w is a divergence-free component that can carry circulation around semantic loops without changing local mass directly.

3. Semantic Modes as Reaction–Diffusion

We evolve semantic modes $\psi(x, t) \in \mathbb{R}^k$ by a reaction–diffusion system:

$$\partial_t \psi = \nabla_{\mathcal{M}} \cdot (K \nabla_{\mathcal{M}} \psi) + R(\psi; \rho, \phi), \quad (9)$$

where K is a positive (semi-)definite diffusion tensor/operator.

A concrete, widely-used example for R is competitive logistic dynamics (component-wise),

$$R_a(\psi; \rho) = r_a(\rho) \psi_a (1 - \psi_a) - \sum_{b \neq a} \kappa_{ab} \psi_a \psi_b, \quad a = 1, \dots, k, \quad (10)$$

with $\kappa_{ab} \geq 0$ and a density-gated growth rate such as

$$r_a(\rho) = \sigma(\rho - \rho_a^*), \quad (11)$$

where $\sigma(\cdot)$ is a smooth step (e.g., logistic).

Coupling back into flow. Beyond the pressure tilt above, ψ can reshape the effective geometry/permeability of the manifold (or graph below), e.g., by modulating weights so that certain modes preferentially open or close semantic channels.

4. Graph-Based Discretization (Manifold $\rightarrow$ Semantic Graph)

In practice, $\mathcal{M}$ is represented by a semantic graph $G = (V, E)$, where vertices are concept/token embeddings and edges encode similarity or learned relational structure. Let $n = |V|$ and $m = |E|$.

4.0 The Incidence Matrix Approach (Implementation-Oriented Discretization)

The key modeling choice is to place transport dynamics on *edges* via an edge flux j , rather than attempting to define a node-wise velocity. This matches a finite-volume viewpoint: node variables such as ρ are conserved quantities, while movement occurs across edges as flux. Thus the continuity update becomes unambiguous and conservative:

$$\dot{\rho} + B^\top j = s_\rho. \quad (12)$$

A practical challenge is the discrete nonlinear advection term $A(j, \rho)$ used in edge-momentum updates (Section 4.4), which typically requires stabilization (e.g., upwinding / controlled numerical diffusion) for robustness.

4.1 Incidence-Matrix Calculus (Node Fields + Edge Fluxes)

Fix an arbitrary orientation of edges and define the (signed) incidence matrix $B \in \mathbb{R}^{m \times n}$: for an edge $e = (i \rightarrow j)$, the row B_e has -1 at column i , $+1$ at column j , and zeros elsewhere.

Let $W_e = \text{diag}(w_e) \in \mathbb{R}^{m \times m}$ be edge weights ($w_e > 0$). Define:

$$\text{Graph gradient: } \nabla_G \phi = B\phi \in \mathbb{R}^m, \quad (13)$$

$$\text{Graph divergence: } \nabla_G \cdot j = B^\top j \in \mathbb{R}^n, \quad (14)$$

$$\text{Graph Laplacian: } L = B^\top W_e B \in \mathbb{R}^{n \times n}. \quad (15)$$

We discretize node fields:

$$\rho(t) \in \mathbb{R}^n, \quad \phi(t) \in \mathbb{R}^n, \quad \psi(t) \in \mathbb{R}^{n \times k}, \quad (16)$$

and treat transport primarily through an *edge flux* $j(t) \in \mathbb{R}^m$.

4.2 Discrete Continuity

A graph finite-volume style continuity equation is:

$$\dot{\rho} + B^\top j = s_\rho(t), \quad (17)$$

where $s_\rho(t) \in \mathbb{R}^n$ is an optional injection/removal term.

4.3 Flux Model: Conservative + Circulatory Components

We define pressure node-wise:

$$p_i(t) = \Pi(\rho_i(t); \psi_i(t)), \quad p(t) \in \mathbb{R}^n. \quad (18)$$

A conservative (potential-driven) flux is

$$j^{(c)} = -W_e B \phi. \quad (19)$$

To allow structured circulation around graph cycles, we add a residual flux $j^{(r)}$:

$$j = j^{(c)} + j^{(r)}. \quad (20)$$

We constrain $j^{(r)}$ to be divergence-free (no mass creation/destruction):

$$B^\top j^{(r)} = 0. \quad (21)$$

When a cycle basis matrix C (cycle-edge incidence) is available, one may measure circulation via

$$\text{curl}_G(j^{(r)}) \equiv C j^{(r)}. \quad (22)$$

4.4 Edge-Momentum (One Practical Discretization)

To keep an explicit inertia term while living in the edge-flux world, treat j as the primary dynamical variable:

$$\dot{j} + A(j, \rho) = -W_e B p - \alpha j + f_e(t), \quad (23)$$

where:

- $A(j, \rho)$ is a discrete nonlinear advection term (implemented, e.g., on the line graph or via upwinded edge-to-edge transport),
- $-W_e B p$ is the discrete pressure-gradient forcing,
- $f_e(t) \in \mathbb{R}^m$ is external edge forcing from inputs,
- $\alpha \geq 0$ is damping.

4.5 Discrete Reaction–Diffusion for Modes

Using the Laplacian L ,

$$\dot{\psi} = -K L \psi + R(\psi; \rho, \phi), \quad (24)$$

where K may be scalar, diagonal, or a full operator acting across mode channels.

Mode-shaped conductance (optional but SOL-flavored). Let edge weights adapt to modes:

$$w_e(\psi) = w_e^{(0)} \exp\left(\gamma^\top \frac{\psi_i + \psi_j}{2}\right) \quad \text{for } e = (i, j), \quad (25)$$

so that active modes open/close semantic pathways.

5. Physics-Informed Neural Network (PINN) Surrogates

We approximate the continuous (or discrete) fields with neural networks:

$$u_\theta(x, t) = (\rho_\theta(x, t), v_\theta(x, t), \phi_\theta(x, t)) \approx u(x, t), \quad (26)$$

$$\psi_\theta(x, t) \approx \psi(x, t). \quad (27)$$

Define PDE residual operators by substituting (u_θ, ψ_θ) into governing equations:

$$r_{\text{flow}}(x, t; \theta) = \mathcal{N}_{\text{flow}}[u_\theta](x, t), \quad (28)$$

$$r_{\text{mode}}(x, t; \theta) = \mathcal{N}_{\text{mode}}[\psi_\theta, u_\theta](x, t), \quad (29)$$

where $\mathcal{N}_{\text{flow}}$ encodes the manifold flow equations and $\mathcal{N}_{\text{mode}}$ encodes the mode dynamics.

(For graph implementations, residuals are defined on nodes/edges using (17)–(23) and the constraint (21).)

6. Learning Objective

6.1 Data Fidelity Loss

Given observations u_{obs} and/or ψ_{obs} ,

$$\mathcal{L}_{\text{data}}(\theta) = \mathbb{E}_{(x,t) \sim \mathcal{D}} \left[\|u_\theta(x, t) - u_{\text{obs}}(x, t)\|^2 + \|\psi_\theta(x, t) - \psi_{\text{obs}}(x, t)\|^2 \right], \quad (30)$$

dropping terms for unobserved components.

6.2 PDE Residual Loss

At collocation points $(x, t) \sim \mathcal{C}$,

$$\mathcal{L}_{\text{PDE}}(\theta) = \mathbb{E}_{(x,t) \sim \mathcal{C}} \left[\|r_{\text{flow}}(x, t; \theta)\|^2 + \|r_{\text{mode}}(x, t; \theta)\|^2 \right]. \quad (31)$$

6.3 Boundary and Initial Conditions

Let $\mathcal{B}[u_\theta](x, t)$ denote boundary/initial operators and $b(x, t)$ prescribed values:

$$\mathcal{L}_{\text{BC}}(\theta) = \mathbb{E}_{(x,t) \sim \mathcal{C}_{\text{BC}}} \left[\|\mathcal{B}[u_\theta](x, t) - b(x, t)\|^2 \right]. \quad (32)$$

6.4 Variational Free Energy (Active Inference / Generative Link)

Introduce latent variables z with approximate posterior $q_\eta(z|x)$ and generative model $p_\theta(x, z)$. Define variational free energy (negative ELBO):

$$\mathcal{L}_{\text{FE}}(\theta, \eta) = \mathbb{E}_{x \sim P_X} \left[\mathbb{E}_{z \sim q_\eta(z|x)} (\log q_\eta(z|x) - \log p_\theta(x, z)) \right], \quad (33)$$

where encoder parameters η are distinct from the scalar potential ϕ .

6.5 Total Objective

$$\mathcal{L}(\theta, \eta) = \mathcal{L}_{\text{data}}(\theta) + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}}(\theta) + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}(\theta) + \lambda_{\text{FE}} \mathcal{L}_{\text{FE}}(\theta, \eta), \quad (34)$$

with nonnegative hyperparameters $\lambda_{\text{PDE}}, \lambda_{\text{BC}}, \lambda_{\text{FE}}$.

7. Boundary-Layer Hotspot Detection

Define a hotspot functional highlighting sharp semantic structure:

$$H(x, t; \theta) = \alpha_1 \|\nabla_{\mathcal{M}} \rho_{\theta}(x, t)\| + \alpha_2 \|\nabla_{\mathcal{M}} \phi_{\theta}(x, t)\| + \alpha_3 \|\text{curl}(w_{\theta})(x, t)\|, \quad (35)$$

with $\alpha_i \geq 0$.

On graphs, replace $\nabla_{\mathcal{M}}$ with $B(\cdot)$ and (optionally) use cycle curl Cj :

$$H_i(t) = \tilde{\alpha}_1 \|(B\rho)_{\text{edges incident to } i}\| + \tilde{\alpha}_2 \|(B\phi)_{\text{edges incident to } i}\| + \tilde{\alpha}_3 \|(Cj)_{\text{cycles touching } i}\|. \quad (36)$$

Points/regions where $H \geq \tau$ for threshold $\tau > 0$ are treated as hotspots: regions where SOL should allocate more attention, refine discretization, or adapt generative behavior.

8. Intellectual Merit: From Static Retrieval to Dynamic Flow

Modern semantic systems (Transformers, RAG) largely operate on static similarity in an embedding space. SOL proposes a shift: semantics as time-dependent transport with inertia, regulation, and structured circulation.

- **Inertia vs. instantaneity:** momentum encourages coherent trajectories rather than brittle jumps.
- **Pressure as regulation:** $\Pi(\rho; \psi)$ offers a physics-based mechanism for crowding control without hard truncation.
- **Structured recurrence:** circulation components capture “looping” conceptual dynamics (recursion, reframing, indirect association) as a first-class signal rather than a bug.

A schematic comparison between static retrieval and SOL-style dynamics is shown in Figure 2.

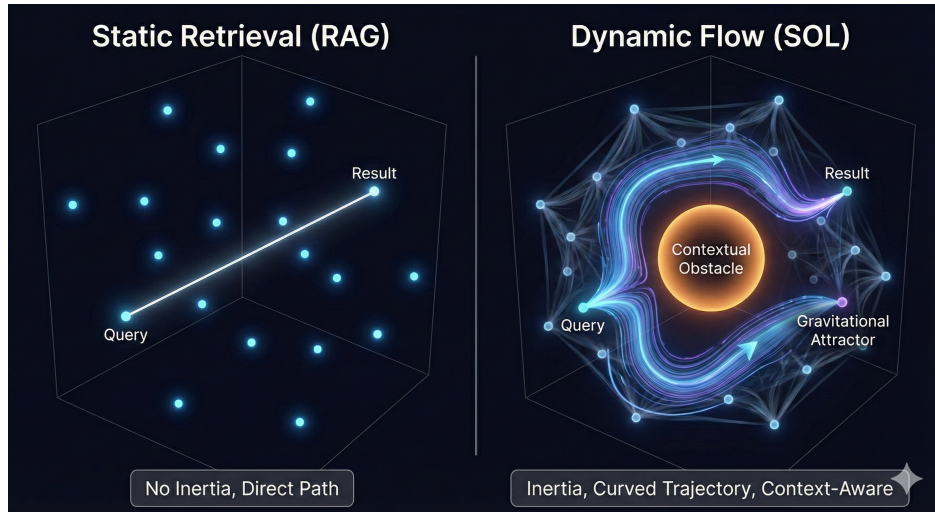


Figure 2: **Comparison of semantic processing paradigms.** **Left (RAG):** Traditional static vector retrieval finds the nearest neighbor via a direct path, treating intermediate context implicitly. **Right (SOL):** The proposed dynamic flow model treats semantics as a fluid with inertia. The path from Query to Result is curved, influenced by the “gravity” of attractor topics and forced to navigate around high-pressure contextual obstacles.

9. Notes on Approximations and Implementation Risks

(i) Curvature and covariant momentum (Section 2). Equation (7) uses a chartwise “Euclidean” inertial term. The fully covariant form replaces $\partial_t v + (v \cdot \nabla_{\mathcal{M}})v$ with $\nabla_{\partial_t} v + \nabla_v v$ (Levi-Civita connection). Curvature corrections are available if $\mathcal{M}$ is not well-approximated by locally flat charts.

(ii) Energy and dissipation (Section 2). Without damping/viscosity, pressure gradients can accelerate flow indefinitely. In practice SOL relies on explicit sinks (e.g., $\alpha > 0$ and/or $\mu > 0$), stable pressure laws, and (in discrete settings) mild numerical diffusion or flux limiting near sharp fronts. A stricter conservative extension would add an energy/entropy equation or use shock-capturing / relaxation forms.

(iii) Interpreting friction α (Section 2). Cognitively, α may be read as forgetting/decay (damping transient motion) or resistance-to-change (stabilizing settled states). A practical refinement is adaptive damping, e.g. $\alpha = \alpha(\rho, \|\nabla_{\mathcal{M}}\phi\|, \text{residual})$ to encourage exploration far from coherent basins and stabilization near solutions.

(iv) Circulatory flux specification (Section 4). The residual flux $j^{(r)}$ should not be a free “escape hatch.” The constraint $B^\top j^{(r)} = 0$ in (21) forces $j^{(r)}$ to live in the cycle space $\ker(B^\top)$ (a discrete analogue of divergence-free flow). A stronger construction is to parameterize $j^{(r)}$ directly via a cycle-space basis or a weighted orthogonal projection.

(v) Cycles at scale (Section 4). Computing a full cycle basis can be expensive on large graphs. For early implementations, circulation can be approximated using fundamental cycles from a spanning tree, or local “curl proxies” based on short cycles (triangles / 4-cycles) discovered via sparse adjacency operations.

(vi) Stabilizing discrete advection (Sections 4.0–4.4). The advection term $A(j, \rho)$ is typically the main numerical instability source. Robust schemes often use semantic upwinding (donor selection by flow sign), controlled numerical diffusion (e.g., local Lax–Friedrichs/Rusanov), and positivity-preserving flux limiting to maintain $\rho \geq 0$.

(vii) PINN training stability (Section 6). Physics residual losses often start much larger than data losses, so naive weighting can ignore physics or destabilize learning. Practical training commonly uses curriculum schedules, adaptive loss balancing (e.g., gradient-norm methods such as GradNorm), and residual-guided collocation sampling (often aligned with hotspots).